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ONLINE LESSON O LEVEL (UCE)

S3 & S4

DATE: **Monday, 24th August 2026**

TIME: 5:00p.m to 6:30p.m today

SUBJECT: PHYSICS

ELEMENT: SECTION B (QUESTION APPROACH)

FACILITATOR: Mr.**NSEREKO ROBERT**

Mode: Google meet

0750661248

Reference Document: Wakissha uce physics 2026.pdf

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ELEMENTS OF CONSTRUCT

Item	Element of construct	Topics covered
Section A(compulsory)		
	Understands how waves are generated, propagated and their application in everyday life	➢ Nature of light; reflection of light at plane surfaces ➢ Reflection of light at curved surfaces ➢ Refraction, dispersion, and colour ➢ Lenses and optical instruments ➢ General wave properties ➢ Sound waves
Item two(2)	Understands the structure of atoms, nuclear processes and their application in every day life.	➢ Nuclear processes ➢ Atomic models
Item three(3)	Understands solar system galaxies, stars, satellites & digital communication in everyday life	➢ The solar system ➢ Stars and galaxies ➢ Satellites and communication ➢ Digital electronics
Section B (part I) Attempt one question only		
Item four and five(4 & 5)	Understands effect of force and heat on	➢ Measurements in Physics

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	properties of matter	➢ States of matter ➢ Effects of forces ➢ Work, energy, and power ➢ Turning effect of forces, centre of gravity, and stability ➢ Pressure in solids and fluids ➢ Mechanical properties of Materials and Hooke's law ➢ Linear and non-linear motion ➢ Temperature measurements ➢ Heat transfer ➢ Expansion of solids, liquids, and gases ➢ Heat quantities and vapours
Section B (part II) Attempt one question only		
Item six and seven (7 & 8)	Appreciates electricity and magnetism in every day life Magnets and magnetic fields	➢ Electromagnetic effects ➢ Electrostatics ➢ Introduction to current electricity ➢ Voltage, resistance and Ohm's law ➢ Electric energy

		distribution and consumption

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FACILITATION ON MECHANICS AND HEAT PROPERTIES

WAKISSHA UCE PHYSICS ITEM 4&5

PREPARED BY: Mr.NSEREKO ROBERT

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Item 4

A group of scouts on a camping trip noticed something peculiar about their flag at the lake shore. During hot days the flag flew towards land and at night, it flew towards sea even when there was no wind. They were puzzled by this observation. Later they met a local farmer who had recently built a dam on his farm. He was too unsure whether to fill the dam to its full capacity and why the walls of the dam were so thick at the bottom.

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Additionally the farmer was concerned about the quality of milk produced on his farm and suspected that his employees might be diluting it.

Hint: the dam is 8 metres deep and can support a maximum pressure of 70,000 Pa.

Density of water = 1000 kgm^{-3}

Task

As a student of physics;

(a) explain to the scouts why the flag behaved that way.

(b) help the farmer understand the purpose of thick walls at the bottom in the dam and advise him on filling the dam to capacity.

(c) suggest a way for the farmer to determine if the milk was diluted or if there were other factors affecting it.

(d) state the applications of the atmospheric pressure.

Response

Item 4

(a) Explanation of the Flag's Behavior (Sea and Land Breezes):

During hot days, the land absorbs heat and warms up much faster than the lake water because water has a higher specific heat capacity. The air directly above the land becomes warm, expands, and rises, creating a localized area of low pressure. Cooler, denser air from over the lake rushes in to replace this rising warm air. This convection current is called a sea breeze, which blows the flag towards the land.

At night, the land loses its heat and cools down faster than the lake. The air

above the relatively warmer water rises, creating low pressure over the lake. Cooler air from the land flows towards the lake to replace it. This is a land breeze, causing the flag to fly towards the sea.

(b) Purpose of Thick Walls at the Bottom of the Dam & Filling Advice:

Purpose of thick walls: In fluids, pressure is directly proportional to depth (Pressure = hρg). As you go deeper into the dam, the hydrostatic pressure exerted by the water increases significantly. The walls at the bottom must be constructed thicker and stronger to withstand this enormous pressure and prevent structural failure or bursting.

Filling Advice (Calculation):

Maximum depth (h) = 8 m

Density of water (ρ) = 1000 kg/m³

Acceleration due to gravity (g) = 10 m/s²

Maximum pressure at the bottom = h × ρ × g

$$= 8 \times 1000 \times 10$$

$$= 80{,}000 \text{ Pa.}$$

The dam can only support a maximum pressure of 70,000 Pa. Because 80,000 Pa is greater than 70,000 Pa, filling the dam to its full 8-meter capacity will exceed the structural limit and cause the dam to break. Therefore, the farmer must NOT fill the dam to capacity.

(c) Determining if the Milk was Diluted:

The farmer should use an instrument called a lactometer (a specialized hydrometer). A lactometer is designed to measure the density (or specific gravity) of milk based on the principle of flotation. Pure milk is denser than water. If employees are diluting the milk with water, the overall density of the milk mixture will decrease. When the lactometer is floated in the diluted milk, it will sink deeper than the calibration mark for pure milk, instantly confirming adulteration.

(d) Applications of Atmospheric Pressure:

- *Drinking liquids using a straw.*
- *Drawing fluids into a medical syringe.*
- *Transferring liquids using a siphon tube.*
- *The operation of lift pumps for drawing well water.*
- *Rubber suckers (suction pads) used for attaching items to smooth surfaces.*

Item 5

In mountainous region, most people stay at the foot of the mountains while others stay at the top. People at the top experience water shortage and nose bleeding while those at the foot experience severe heat during dry spell. An NGO, a voluntary organization wants to help the people reduce on water shortage by installing a solar pump to provide constant water supply of 300 litres every minute but they do not know the height of the mountain. They have contacted a meteologist who gave them a barometer to help them in this task.

Hint:

- Barometer gave that pressure difference between the top and bottom as 10 Kpa.
- Density of air is 1.25 kgm^{-3}.
- Acceleration due to gravity is 10 ms^{-2}.

Task

As a physics student, help the NGO and settlers to:

(a) explain why it is hotter at the bottom and cooler at the top.

(b) explain the causes of nose bleeding and water shortage at the top of the mountain.

(c) estimate the height of the mountain.

(d) determine the power of the pump required to raise water to the top of the mountain.

Response

Item 5

(a) Why it is Hotter at the Bottom and Cooler at the Top of the Mountain:

The Earth's surface is heated by radiation from the Sun. The hot surface heats the air near it. The heated air expands, becomes less dense and rises by convection. As the air rises, atmospheric pressure decreases, causing the air to expand and cool. Hence, it is generally hotter near the Earth's surface and cooler at higher altitudes.

(b) Causes of Nose Bleeding and Water Shortage at the Top:

• ***Nose bleeding****: Atmospheric pressure decreases significantly with increasing altitude. At the top of the mountain, the external atmospheric pressure is much lower than the internal blood pressure pumping through the human body. This large pressure difference exerts an outward force that can cause the delicate, thin-walled capillaries in the nose to rupture and bleed.*

• ***Water shortage****: Gravity pulls surface and ground water downwards towards the lowest possible elevation, meaning the natural water table is situated far below the mountain peak. Moving water up to the top requires tremendous energy to overcome this gravitational pull, making it scarce*

without specialized pumping infrastructure.

(c) Estimating the Height of the Mountain:

Data given:

Pressure difference (ΔP) = 10 kPa = 10,000 Pa

Density of air (ρ) = 1.25 kg/m³

Acceleration due to gravity (g) = 10 m/s²

Using the fluid pressure formula: ΔP = h × ρ × g

10,000 = h × 1.25 × 10

10,000 = 12.5h

$h = \frac{10{,}000}{12.5}$ *= 800 meters.*

The estimated height of the mountain is 800 meters.

(d) Determining the Power of the Pump:

Data given:

Volume of water = 300 liters per minute

Height (h) = 800 m

Time (t) = 1 minute = 60 seconds

Density of water = 1000 kg/m³ (since 1 m³ = 1000 liters, 300 liters is 0.3 m³. Mass = 0.3 × 1000 = 300 kg. Alternatively, 1 liter of water has a mass of 1 kg, so 300 liters = 300 kg).

1. Calculate the Work Done (Potential Energy gained by the water):

Work Done = m × g × h

Work = 300 kg × 10 m/s² × 800 m

Work = 2,400,000 Joules

2. Calculate the Power of the Pump:

$Power = \frac{Work\ Done}{Time}$

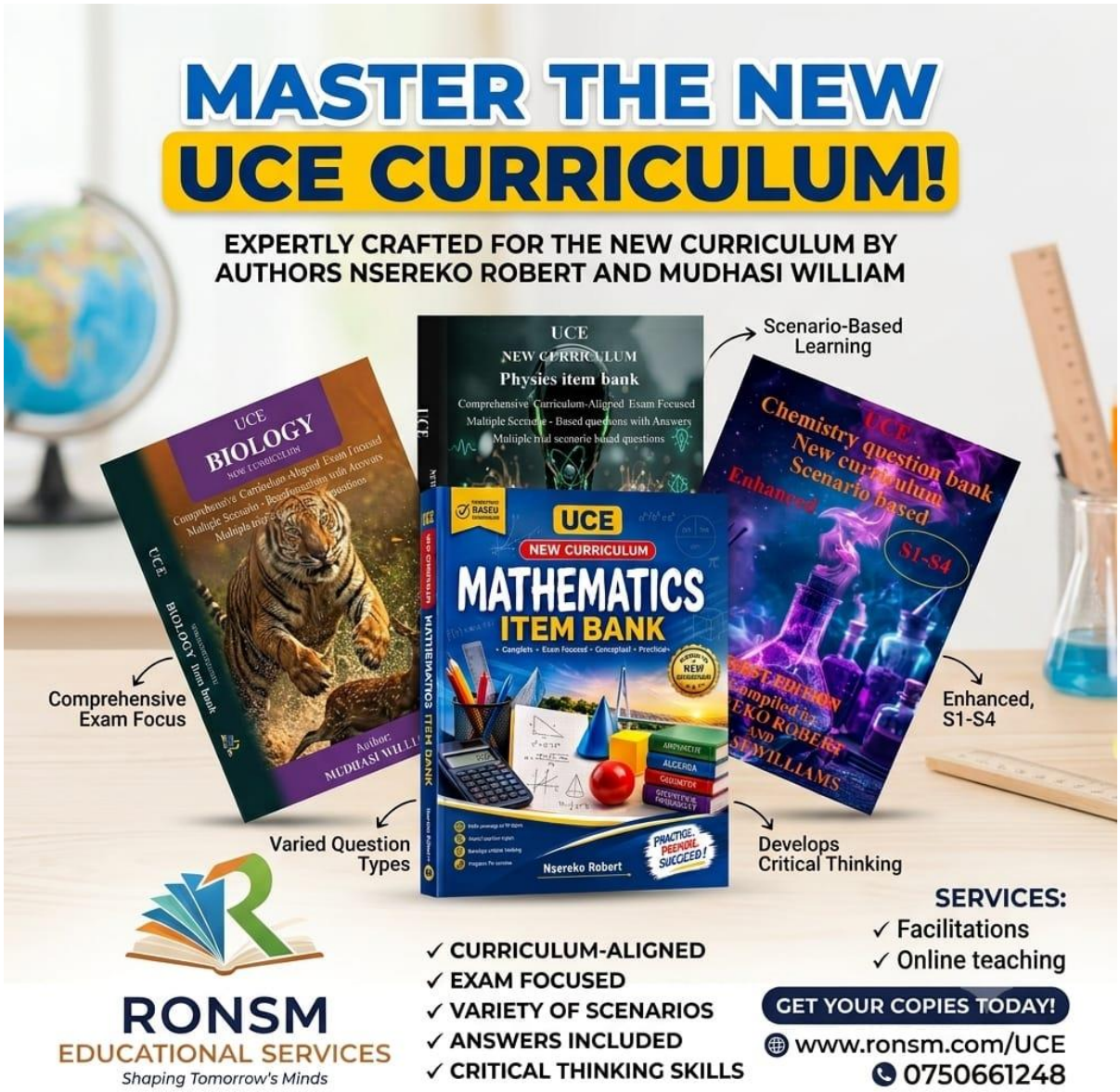
$Power = \frac{2{,}400{,}000\ J}{60\ s}$

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Power = 40,000 Watts (or 40 kW)

The solar pump requires a power output of 40,000 W to successfully raise the water.



Revision paper

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